

Dharanidharan R S

 dharanidharan-R-S |  s2007dharanidharan-cloud |  s2007dharanidharan@gmail.com | +91 8610950296

PROFILE

Mechanical Engineering student with hands-on experience in CAD modeling, automation systems, and product design. Skilled in CATIA, Creo, SolidWorks, ANSYS, and MATLAB, with proven success in engineering competitions and prototype development.

SKILLS

Design & Simulation	CATIA, Creo, SolidWorks, AutoCAD, ANSYS, MATLAB
Programming	C, Python, Java
Areas of Interest	Mechanical Design, Simulation & Analysis, Automation, Renewable Energy Systems
Soft Skills	Problem Solving, Teamwork, Leadership, Communication, Time Management

PROJECTS

PTI Zero Fall Initiative Safety Automation / Design

- Designed a sensor-based safety system with a retractable safety plate to prevent accidents at the Platform-Train Interface (PTI).
- Enhanced passenger safety and railway reliability by automatically reducing the risk of falls between the train and platform.

Automatic Chapathi Making Machine Hardware / Automation

- Designed an automated chapathi-making system integrating conveyor belts, sensors, and mechanical actuation.
- Implemented sensor-based dough detection and rack-and-pinion pressing for uniform chapathi production.

Twin-Turbo V6 Engine CAD Assembly PTC Creo

- Designed and assembled a detailed Twin-Turbo V6 Engine model using PTC Creo.
- Enhanced understanding of engine architecture, component integration, and mechanical design principles.

EDUCATION

2024 – 2028	B.E. Mechanical Engineering, Kongu Engineering College	(CGPA: 8.78/10.0)
2024	Class XII (CBSE), PM SHRI Kendriya Vidyalaya No. 1, Madurai	(85.6%)
2022	Class X (CBSE), PM SHRI Kendriya Vidyalaya No. 1, Madurai	(74.6%)

IN-PLANT TRAINING

JK Fenner (India) Limited Madurai, Tamil Nadu Jul 2025

- Gained practical, hands-on exposure to industrial workflows and operational manufacturing processes.
- Observed and analyzed mechanical systems to bridge the gap between theoretical coursework and real-world industrial applications.

ACHIEVEMENTS

- Winner:** Intercollege Mini Hackathon 2025 for the design and fabrication of an **Automatic Chapathi Pressing Machine** featuring automated dough detection and pressing mechanisms.
- Winner:** Intercollege Ideathon 2025 for proposing the **PTI Zero Fall Initiative**, an automated safety system to prevent passenger accidents at the Platform-Train Interface (PTI).
- Winner:** Paper Presentation & Project Expo at **EKNA 2025**, Anna University, for developing and presenting the **PTI Zero Fall Initiative**; subsequently recognized by **Kongu Engineering College** during the **Annual Day Awards Ceremony**.

TRAINING & CERTIFICATIONS

- NPTEL Online Certification:** Manufacturing Processes – Casting and Joining, IIT Kanpur
- NPTEL Online Certification:** Laser Based Manufacturing, IIT Guwahati
- Non-Formal Course:** EV Design and Simulation using MATLAB
- Non-Formal Course:** CNC Programming
- Certification Course:** ProSimulator